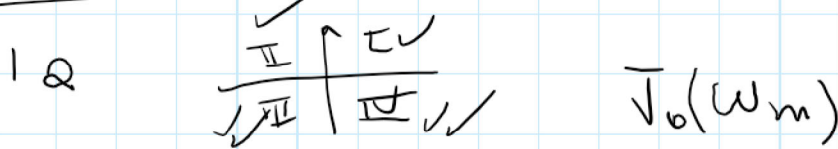


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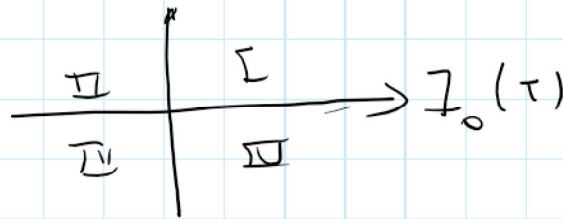
Tuesday, June 9, 2026

8:56 AM

Recall AC/DC choppers



2Q choppers



basic principles

- ① a pair of switch-diode is required for operation in each quarter.
- ② in motoring, always the switch ^{connects} ~~transfers~~ the input voltage to the output, the diode makes a short circuit.
- ③ in braking, is the other way around.
- ④ in modes II, IV, back EMF is negative.

2Q chopper (Q_1 & Q_2)



Q1 operation (motor)

S_1 chops

S_2 always off

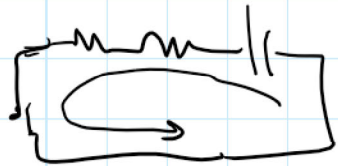
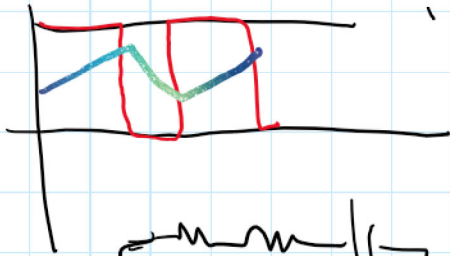
$S_1, on \Rightarrow V_o = V_i$

S_1, off

$S_1, off \quad V_o = 0$

$$\bar{V}_o = D_s V_i, \quad M = \frac{\bar{V}_o}{J} = D_s$$

$$V_o = D_{s1} V_i, M = \frac{V_o}{V_i} = D_{s1} \quad \begin{cases} s_1 \text{ off} \\ D_1 \text{ on} \end{cases} \quad V_o = \phi$$

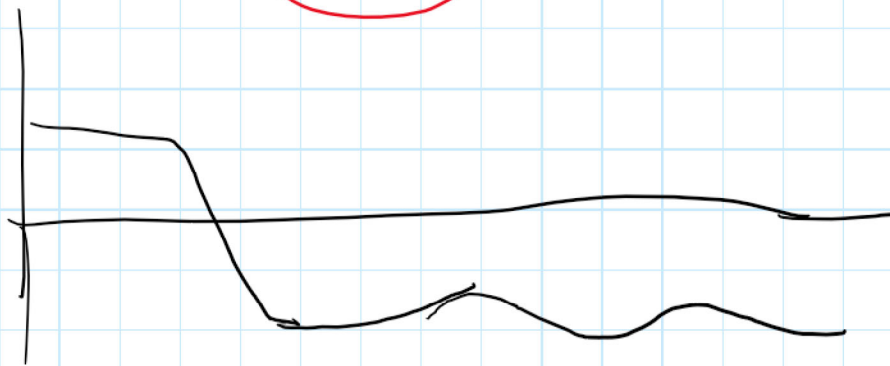


Q_2 operation (braking)

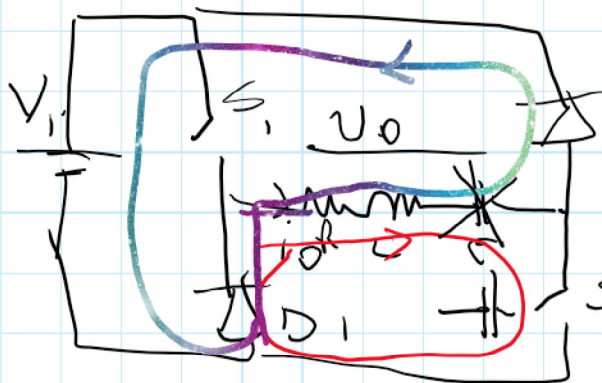
$$\begin{cases} s_1 \text{ off} \\ s_2 \text{ chops} \end{cases} \quad \begin{cases} s_2 \text{ on, } |i_o| \uparrow \\ D_2 \text{ off} \end{cases} \quad V_o = \phi$$

$$\begin{cases} s_2 \text{ off} \\ D_2 \text{ on} \end{cases} \quad \begin{cases} |i_o| \downarrow \\ V_o = V_i \end{cases}$$

$$\bar{V}_o = (1 - D_{s1}) V_i, M = 1 - D_{s1}$$



2Q Chopper (E, IV)



Q_1 operation: (motoring)

s_4 always on

s_1 chops

$s_{1on} \rightarrow V_o = V_i$
 $D_1 \text{ off}$

$$\bar{V}_o = D_{s1} V_i$$

$$M = D_{s1}$$

$s_1 \text{ off} \rightarrow V_o = \phi$
 $D_1 \text{ on}$



$$V_o = D_{S1} V_i$$

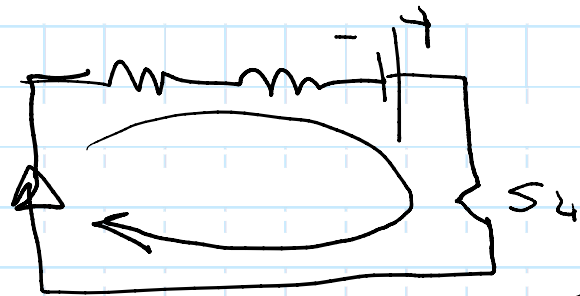
$$M = D_{S1}$$

$$S_1 \text{ off } \rightarrow V_o = 0$$

$$D_1 \text{ on}$$

S_1 always off
 D_1 always conducts

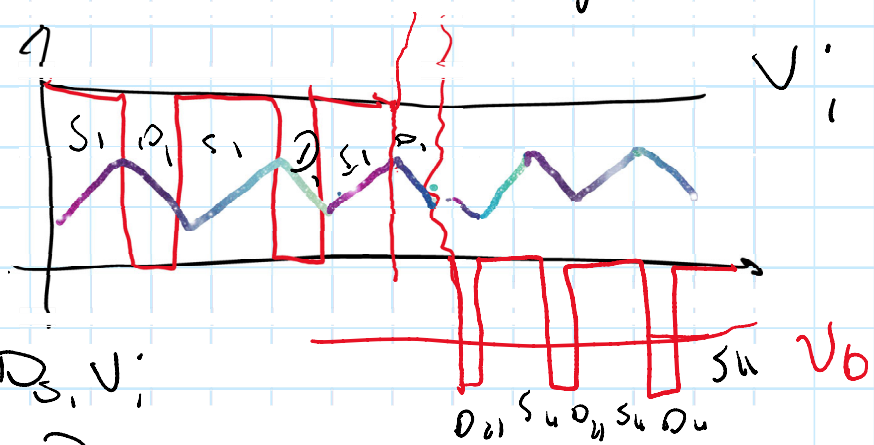
Q4 operation: (braking)
 S_4 chops



S_{4on} $|V_o| \uparrow$
 $D_4 \text{ off}$ $V_o = 0$
 $S_4 \text{ off}$ $|V_o| \downarrow$
 $D_4 \text{ on}$ $V_o = V_i$

$$\tau = K \Phi \omega_m$$

if direction must change.



Q1 $\begin{cases} V_o = D_{S1} V_i \\ M = D_{S1} \end{cases}$

Q4 $\begin{cases} |V_o| = (1 - D_{S4}) V_i \\ M = -(1 - D_{S4}) \end{cases}$



$$I = \frac{100 - 98}{R_a} = I_{nom}$$

$$I = \frac{100 + 98}{R_a}$$



$$I = \frac{100 + 70}{R_a}$$